

Review: “Bayesian Semi-Blind Deconvolution at Scale”

September 15, 2026

1 Summary

This manuscript focuses on a Bayesian model for *semi-blind* de-convolution described in [Senn et al. \(2025\)](#). This is a challenging task where not only the true image is unknown but also the convolution filter. By exploiting conjugacy relations between the various prior and likelihood components, together with the focus on cyclic lattices, the posterior can be efficiently explored through a Gibbs sampler that relies heavily on the use of the fast Fourier transform (FFT) and its inverse (IFFT).

From the computational perspective, this manuscript presents two improvements over [Senn et al. \(2025\)](#): an increased use of the FFT, and an alternative update of the blur that uses a marginal conditional instead of the usual full conditional. According to the experimental results, there is some evidence that the marginal version helps in dealing with multimodality issues that arise when there is little data available.

While I think the subject of the manuscript is interesting and relevant to JCGS, in my opinion the two improvements described above are not important enough to justify publishing a new paper. I expand on this and other issues in the following comments. Excerpts from the manuscript are *emphasized*.

2 Major comments

As I mentioned above, the main issue with this manuscript is the lack of substantial work that is not already found in [Senn et al. \(2025\)](#). A side-by-side comparison shows that almost everything in the paper up to Section 4.1 can be found in the 2025 paper (Section 4.2 describes the marginal HMC sampler which is new). If there are new contributions up to Section 4.1, they are hard to detect because the bulk of the text is dedicated to explaining things that are already in [Senn et al. \(2025\)](#). A new version of the present manuscript should avoid repetition of previous work as much as possible, and anything that must be included should be placed in Section 2 (preliminaries) so that readers can clearly distinguish the novel contributions from past work. Moreover, a new

version should at the very least attempt a more thorough empirical examination of the pros and cons of the two competing samplers.

On a more technical note, I think there may be a theoretical problem with the way the Gaussian field is defined on the cyclic lattice. Following [Borovitskiy et al. \(2020\)](#), the so-called *naive generalization* of a Gaussian process in Euclidean space to a compact manifold, where the Euclidean norm is simply substituted with the geodesic distance, does not yield a positive definite kernel. But this is precisely what [Senn et al. \(2025\)](#) and the present manuscript do to build the Gaussian field on the flat torus (i.e., the cyclic lattice). In a practical numerical implementation, this would show up as negative eigenvalues that need to be truncated to 0 in order to avoid taking the square root of a negative number (see line 2 in Algorithm 2 of the manuscript).

Interestingly, [Borovitskiy et al. \(2020\)](#) show that for the flat torus, the naive generalization is correct to a first order approximation, which means that for short length scales and/or large embedding grids the resulting covariance matrices may be positive definite. This is the same phenomenon that underlies the circulant embedding technique for sampling general stationary Gaussian random fields ([Graham et al., 2018](#)). Thus, when the authors justify the use of an embedding lattice to remove boundary effects, they should also mention that a larger embedding is needed to handle non-definiteness issues that may show with sufficiently large length scales.

An alternative that completely elides the non-definiteness issue is to work with the conditional autoregressive model $\text{CAR}(\rho, \sigma^2)$ with $\rho \in [0, 1)$, which yields positive definite precision—and thus covariance—matrices regardless of the graph being considered ([Gelfand and Vounatsou, 2003](#), Section 3).

3 Minor comments

Section 1

Comment 1: P.1 L9–10: “...reflectivity image, well reflectivity, noise and seismic wavelet...”. These terms are not known in the statistical community. They should first be explained before being discussed.

Comment 2: P.4 L50–51: book citations need to include a pointer to the section that contains the claim; e.g., (Liu, 2008, Section X.Y.Z). This happens multiple times in the manuscript, with Liu’s book but also with Davis’ and Rue & Held.

Section 2

Comment 3: P.5 L39–40: using $p(x)$ to denote the density of a parametrized family like $\mathcal{N}_n(\mu, \Sigma)$ is bad practice, because there are multiple dimensions, means, and covariance parameters considered in this study. For example, a fairly

common approach is to write $\mathcal{N}_n(x; \mu, \Sigma)$ for such a density; many other options are possible.

Comment 4: P.5 L46–49: consider adding a citation for these formulas or a short proof. I think that statisticians are more aware of the case of a multivariate normal vector $z = (z_1, z_2)$ and the conditional distribution of z_2 given z_1 is conditioned on either of its components (see e.g. [Mardia et al., 1979](#), Thm 3.2.4), which in turn can be used to get Eq (3).

Comment 5: P.5 L56–57: the error does not enter the formula for the covariance matrix so the value of the error cannot affect the rank.

Comment 6: P.6 L15–16: cite [Rue and Held \(2005, Section 2.3.3\)](#)

Comment 7: Eq 7: The matrices F_n and F_n^H are interchanged; see [Davis \(1979, Eq. 3.2.7\)](#). Also happens in P.8 L14–15.

Comment 8: P. 8 L45–46: formula for the mn eigenvalues is wrong, should be $\lambda_C = \sqrt{mn}(F_m \otimes F_n)^H \text{vec}(\Pi)$.

Comment 9: Algorithm 2: this is essentially Algorithm 2.10 in [Rue and Held \(2005\)](#), cite it. Note that page 65 of that book states clearly that the assumption is that the precision (and thus the covariance) matrix is symmetric positive definite.

Section 3

Comment 10: Most of this section is from [Senn et al. \(2025\)](#), so it should go in Section 2 if absolutely necessary.

Section 4

Comment 11: Section 4.1 is mostly all in [Senn et al. \(2025\)](#), so it should be placed in Section 2 if absolutely necessary.

Section 5

Comment 12: Figure 2: I think this figure is nice but it doesn't quite convey the message that the HMC version is better at handling the multimodality. It would be nice have another plot that focused only on the most volatile index of the blur vector so that a traditional histogram could be shown. A sampler that truly overcomes the multimodality issue would have roughly half of its samples on each mode. This fact cannot be easily assessed from Fig 2.

Comment 13: P. 28 L21–22: typo, should be “lattice”.

Comment 14: Section 5.2: Would it be possible to do a similar analysis as in the previous section where some of the exact observations are withheld? This would help showcase the benefits of the HMC version.

References

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